BALA SHUNMUGAM M

Electrical Engineer + MBA Operations · Plant Floor to P&L

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PROFILE

Electrical & Electronics Engineer pursuing MBA in Operations & Business Analytics. Combines plant-floor technical depth (gas turbine systems, protection relays, motor energy audits at VGTPS) with structured operations frameworks (Lean, Six Sigma, EOQ, TPM, NPV). Built two working browser-based demonstrator tools — an integrated inventory control system (EOQ+ROP+Safety Stock+ABC) and a Six Sigma DMAIC workbench. Looking for an operations or analytics role bridging electrical systems and business outcomes.

EDUCATION

MBA — Operations & Business Analytics *2025 – 2027*
Amrita School of Business, Amrita Vishwa Vidyapeetham, Coimbatore

B.E. — Electrical & Electronics Engineering (CGPA 7.12) *2019 – 2025*
PSNA College of Engineering and Technology, Dindigul

Higher Secondary (CBSE) — 7.7 *2017 – 2019*
Velammal Vidyalaya Viraganoor, Madurai

FEATURED PROJECTS

Inventory Control System for Small Retail — EOQ + ROP + Safety Stock + ABC *MBA Applied Concept · Live demonstrator*

- Designed an integrated inventory control concept for unorganised small retail; demonstrated with a 12-SKU realistic basket modelled on a typical Madurai grocery shop.
- Integrated four classical frameworks — EOQ, Reorder Point, Safety Stock, and ABC analysis — into a single browser-based decision engine with editable inputs.
- Shipped a working demonstrator tool (InventoryIQ) with manager dashboard + mobile-first UI; owned the spec, framework integration and UX design (implementation AI-accelerated).

Operations Study of a Combined-Cycle Gas Power Plant — VGTPS, TANGEDCO *MBA Operations · Field study from in-plant training*

- Operations management study of TANGEDCO's 187.2 MW combined-cycle plant (Phase I GE Frame 6FA + Phase II Ansaldo), built from VGTPS plant documents collected during February 2025 in-plant training.
- Applied TANGEDCO's own five operational lenses (O&M, Crisis Management, Safety, Reliability, Efficiency) to multi-year performance data — including the maintenance-deferral pattern that preceded the May 2010 and June 2012 forced outages.
- Built SWOT and phased improvement roadmap; identified vibration analytics on rotating equipment as the highest-impact reliability lever, given rotating-equipment failures drove most major outages.

Adaptive Protection Relay Coordination for Gas Turbine Generator — VGTPS *B.E. Final Year · Protection coordination study*

- Designed an adaptive 4-group 87G differential relay scheme for a GE Frame 6FA generator using real VGTPS plant data: 51,921 A bus fault level, 172 V stability voltage, 4,429 A rated current.
- Analytical coordination across 10 fault scenarios in 5 protection zones (87G · 40 · 46 · 21 · 87B); demonstrated via relay characteristic math how adaptive settings detect high-resistance stator faults that fixed settings miss.
- Directly motivated by the real May 2010 Valuthur stator earth fault that caused a 3-month outage, learned about during in-plant training. Grounded in IEEE C37.102 and IEC 60909.

Motor Energy Audit Fieldwork — VGTPS Phase I *B.E. In-Plant Training · Mentor-guided EEE fieldwork*

- Hands-on energy-audit fieldwork at TANGEDCO's Valuthur Gas Turbine Power Station Phase I during February 2025 in-plant training; personally took clamp-meter and power-analyser readings on an 8-motor sample of auxiliary drives (cooling-water pumps, fans, compressors, lube/feed pumps).
- Found a recurring load-factor pattern: 3 motors under-loaded at 33–42% (PF 0.55–0.61), 2 motors overloaded at 110–112% rated kW, 3 motors well-matched at 76–86% — informing VFD-retrofit prioritisation.
- Plant subsequently analysed the data to project ~20% aux-electricity reduction with ~14-month VFD payback on the audited motors (financial figures from plant's analysis, not mine). Phase I aux consumption is 5.55–6.60% of total generation per plant 5-year data.

Sponge Iron Operations Study — Agni Steels *MBA Internship · 100 TPD DRI plant, Perundurai*

- Observational operations study at Agni Steels' coal-based rotary-kiln sponge iron division (100 TPD ≈ 36,000 TPA capacity).
- Mapped the full DRI process flow (raw materials → kiln → cooler → magnetic separation → storage); applied Pareto + Fishbone analysis to operator-reported issues.
- Identified kiln accretion (ring formation) and raw material variability as the dominant constraints — proposed predictive ANN soft-sensor as a future improvement path.

MBA BUSINESS ANALYTICS PROJECTS

Customer Segmentation — k-means & Hierarchical Clustering *MBA · Introduction to Business Analytics · R · Quarto · Aug 2025*

- Segmented 30 KTC Company customers by Age, Income, Loan, and Mortgage using Ward.D hierarchical clustering and k-means; elbow method and dendrogram independently confirmed k=3 optimal clusters.
- Built as a complete reproducible Quarto report (R); published on MBA blog. Source, data (DemoKTC.xlsx), and rendered HTML available at portfolio.

Customer Review Text Mining — Cosine Similarity & Document Clustering *MBA · Introduction to Business Analytics · R · tm + slam · Aug 2025*

- Applied full NLP pipeline on 21 restaurant reviews: VCorpus construction → text cleaning → Document-Term Matrix (185 terms, 93% sparsity) → cosine-similarity matrix → Ward.D2 document clustering → word cloud.
- Dominant themes: pizza, good, food, great — with minority negative terms (bad, lousy, slow, filthy). Published on MBA blog with complete R source and reviews.csv data.

Market Basket Analysis — Apriori Association Rules *MBA · Introduction to Business Analytics · R · arules · Aug 2025*

- Ran Apriori algorithm on 38,765 Kaggle grocery transaction records (3,898 members); reshaped long-format CSV into transaction list grouped by Member + Date.
- Key rules: {artificial sweetener} → {whole milk} at 27.58% confidence; {brandy} → {whole milk} at 34.21% confidence (both with low support <0.001 — niche patterns, not dominant trends). Published on MBA blog with PDF and Quarto source.

Consumer Shopping Behavior Analysis — 8-Method Multivariate Study *MBA · Data Analytics using R & Python · Python · Jupyter · 2025*

- Analysed 3,900 retail transactions across 16 variables using 8 methods in Python: T-Test (gender vs spend, p=0.381), ANOVA (seasonal spend, F=3.746, p=0.011), Chi-Square (gender–subscription, χ^2=676.8, p<0.001), PCA, EFA, K-Means clustering, Logistic Regression, and LDA.
- Logistic Regression achieved AUC-ROC = 0.893; discount usage was the dominant subscription predictor (coef 3.37). K-Means (k=3) segmented customers into Young Explorers, High-Value Veterans ($81 avg spend) and Budget Loyals ($39 avg). Jupyter notebook fully reproducible.

TOOLS BUILT

- **InventoryIQ** — browser-based inventory control demonstrator (manager dashboard + mobile UI). Live in-browser EOQ/ROP/Safety-Stock calculation, ABC classification, linear-regression demand forecast, CSV export. Designed the spec and UX; implementation AI-accelerated.
- **Six Sigma Workbench** — DMAIC / DOE / MSA project manager. Built to manage multiple improvement projects with structured tollgates, design-of-experiments setup, and measurement-system analysis.

TECHNICAL SKILLS

Operations & Analytics: EOQ / ROP / Safety Stock, ABC Analysis, Lean (7 Wastes, VSM, Kanban), Six Sigma (DMAIC, DOE, MSA), TPM, OEE, Pareto & Root Cause, NPV / ROI

Data Analytics (R + Python): R: Clustering (k-means, hclust, Ward.D), Text Mining (tm, slam, wordcloud), Association Rules (arules, Apriori), ggplot2, Quarto, dplyr, tidyverse. Python: pandas, numpy, scipy (T-test, ANOVA, Chi-Square), sklearn (PCA, EFA, KMeans, LogisticRegression, LDA), seaborn, matplotlib, Jupyter

Electrical Engineering: Protection relays (87G, 87B, 21, 40, 46), Gas turbine systems (Brayton + Rankine combined cycle, GE Frame 6FA), Motor load analysis & VFD sizing, SCADA exposure, Power evacuation

Tools & Software: MS Excel, MS Project, R (RStudio, Quarto), Python (pandas, scipy, sklearn, seaborn, Jupyter), AutoCAD 2D, HTML/CSS/JS (tool building, AI-assisted)

INDUSTRIAL EXPOSURE

Valuthur Gas Turbine Power Station (VGTPS) — TANGEDCO *Feb 16–21, 2025*

- Natural-gas 187 MW combined-cycle plant (Phase I GE Frame 6FA + Phase II Ansaldo): turbine startup/shutdown sequencing, generator operations, protection systems, power evacuation, auxiliary equipment management.

Subload Dispatch Centre, Pasumalai — Madurai *Feb 10–14, 2025*

- Power distribution operations: load scheduling, dispatch coordination, SCADA-based real-time monitoring, operational risk management in grid operations.

CERTIFICATIONS

- AutoCAD 2D — Internshala (2025)
- MS Excel — Microsoft
- MS Project 2021 — Microsoft